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Al₂O₃-Based a-IGZO Schottky Diodes for Temperature Sensing

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Abstract: High-temperature electronic devices and sensors that operate in harsh environments, especially high-temperature environments, have attracted widespread attention. An Al₂O₃ based a-IGZO (amorphous indium-gallium-zinc-oxide) Schottky diode sensor is proposed. The diodes are tested at 21–400 °C, and the design and fabrication process of the Schottky diodes and the testing methods are introduced. Herein, a series of factors influencing diode performance are studied to obtain the relationship between diode ideal factor n , the barrier height ϕ_B , and temperature. The sensitivity of the diode sensors is 0.81 mV/°C, 1.37 mV/°C, and 1.59 mV/°C when the forward current density of the diode is 1×10^{-5} A/cm², 1×10^{-4} A/cm², and 1×10^{-3} A/cm², respectively.

Keywords: high-temperature; Schottky diodes; Al₂O₃; a-IGZO

1. Introduction

A harsh environment usually involves factors such as extreme temperature, high pressure, and high shock. High-temperature electronics and sensors that can operate between 21 °C and 400 °C have drawn considerable attention owing to their wide applications in harsh environments. Integrated sensing modules that can operate at high temperatures will be beneficial for many industrial applications. A high-temperature integrated circuit is an important part of such systems, so it must comprise diodes and transistors that can operate at high temperatures. The simplest device that can be integrated with a circuit is based on semiconductor diodes [1].

It has already been proved that the application of some wide-bandgap semiconductor materials overcome the defects of silicon materials, making semiconductor devices capable of operating at temperatures above 300 °C [2]. Shao et al. [3] prepared a 4H-SiC-based P-N junction diode using Ni/Ti and Al/Ti/Ni, respectively, with N- and P- 4H-SiC forming an ohmic contact with a forward conduction voltage of 2.6 V at room temperature and 1.4 V at 500 °C at an average voltage drift rate of 2.2 mV/K. Dipalo et al. describe a novel ISFET structure monolithically integrated with an InAlN/GaN HEMT structure [4]. Pearton et al. prepared a sensor based on GaN and can operate at 400 °C [5]. In addition to SiC, gallium nitride (GaN) is also widely used at harsh environments [6–9]. Gregory et al., [10] at the University of Rhode Island in the United States, proposed a thin-film temperature sensor based on a Schottky barrier diode that can be used to test the surface temperature of an ultra-high temperature blade at 25–1100 °C. Among the available oxide semiconductor films, amorphous indium-gallium-zinc-oxide (a-IGZO) is a preferred material for